

Saturated Superfluid Vacuum VI-a

Galactic-Scale Resonances: The Holographic Sheet
as a Coherent Soliton in Ψ

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Abstract

We extend the saturated superfluid vacuum framework of Papers I–III to galactic scales. The central result is that a spiral galaxy disc is a coherent *planar soliton* in the macroscopic order parameter Ψ , phase-locked to the standing wave emitted by the central black-hole breather at eigenfrequency $f_{\text{BH}} = f_p (m_p/M_{\text{BH}})$, where $f_p = m_p c^2/h$ is the proton Compton frequency. Beyond an inner aggregation core r_0 , the soliton produces a Mestel surface density $\Sigma(r) = \Sigma_0(r_0/r)$, which yields exactly flat rotation curves from baryonic matter alone—no dark matter halo is required or permitted. The characteristic node spacing $\Delta r = \mathcal{C}/M_{\text{BH}}$, with $\mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_\odot$, is anchored to Andromeda ($\Delta r \approx 12.9 \text{ kpc}$) and reproduces the quasi-periodic wiggles observed in high-resolution rotation curves. Gravitational coupling is universal because the standing wave operates through spacetime curvature alone—it slows time, and all matter follows geodesics. Applied to the Andromeda system, the framework predicts that the observed razor-thin plane of ~ 30 co-rotating dwarf satellites is the nodal plane of the standing wave, with thickness set by the hydrodynamic de Broglie scale of the medium rather than dark-matter subhalo statistics. We fit the high-resolution NGC 224 rotation curve (625 points, 0.05 kpc steps) and obtain RMS residual $\sim 7.5 \text{ km/s}$ versus $\sim 27 \text{ km/s}$ for ΛCDM at equal parameter count. The dark energy term Λ is absent from this model; the phenomena it was introduced to explain are reinterpreted as emergent hydrodynamic effects.

Contents

1 Introduction

Papers I–III of this series established: (I) fundamental particles as stable topological defects in a saturated superfluid vacuum, with masses organised on an α -harmonic ladder [?]; (II) all four forces as pressure-gradient modes of the same medium, with gravity identified as the acoustic Bjerknes force between vibrating breather modes (Paper II, §3.4) [?]; (III) time as the irreversible evolution of the hydrodynamic wake, with entropy measured by turbulent complexity [?].

The present paper extends the framework to its most directly observable large-scale consequence: the structure of galaxies. In standard cosmology, the flat rotation curves of spiral galaxies are taken as primary evidence for dark matter haloes. We show that no such haloes are needed. The same order parameter Ψ that produces particles as localised vortex defects supports large-amplitude, long-wavelength *coherent* solutions on galactic scales — solitonic standing waves anchored by the central black hole. These waves gravitationally couple to all matter through time dilation alone, producing flat rotation curves as a hydrodynamic consequence of the medium’s equation of state.

The structure of the paper is as follows. Section 2 derives the black-hole eigenfrequency and the spatial standing wave it imprints across the disc. Section ?? establishes the planar soliton as a solution to the logarithmic Schrödinger equation (LogSE) and derives the Mestel surface density profile. Section ?? explains how the soliton couples gravitationally to all forms of matter through time dilation. Section ?? applies the framework to Andromeda: the rotation curve fit and the plane of satellites. Section ?? addresses the absence of dark energy. Section ?? states testable predictions. Section ?? concludes.

2 Black-Hole Eigenfrequency and the Standing Wave

2.1 The Central Breather

In Paper II [?], the proton is identified as a spherical breather mode of the superfluid vacuum (Paper I, §3.2) [?] — a periodically oscillating, localised density perturbation stabilised by a trefoil vortex skeleton. Its characteristic frequency is

$$f_p = \frac{m_p c^2}{h} \approx 2.27 \times 10^{23} \text{ Hz}, \quad (1)$$

the proton Compton frequency. A supermassive central black hole is, in this framework, a macroscopic breather condensate of mass M_{BH} . By the universal frequency–mass relation $f \propto 1/M$ (a heavier condensate circulates more slowly), its eigenfrequency is

$$\boxed{f_{\text{BH}} = f_p \frac{m_p}{M_{\text{BH}}} = \frac{m_p^2 c^2}{h M_{\text{BH}}}} \quad (2)$$

The corresponding de Broglie wavelength of the condensate, propagating at c , defines a characteristic spatial scale:

$$\lambda_{\text{BH}} = \frac{c}{f_{\text{BH}}} = \frac{h M_{\text{BH}}}{m_p^2 c}. \quad (3)$$

2.2 Spatial Standing Wave and Wiggle Spacing

The central breather radiates a cylindrical standing wave across the galactic disc. In cylindrical coordinates (r, ϕ, z) with azimuthal symmetry, the lowest eigenfunction of the Laplacian is the zeroth-order Bessel function:

$$\Psi_{\text{wave}}(r) = J_0(kr), \quad k = \frac{\pi}{\Delta r}, \quad (4)$$

where Δr is the node spacing. Matching to the de Broglie scale and calibrating to the observed wiggle structure in Andromeda’s rotation curve ($\Delta r \approx 12.9$ kpc for $M_{\text{BH}} = 1.4 \times 10^8 M_{\odot}$) yields

$$\Delta r = \frac{\mathcal{C}}{M_{\text{BH}}}, \quad \mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}. \quad (5)$$

The calibration constant $\mathcal{C}$ is currently anchored to a single galaxy; Section ?? constrains it using a five-galaxy sample and outlines the first-principles BdG derivation path that should eventually eliminate it as a free parameter.

For large r , the Bessel function decays as $J_0(kr) \sim \sqrt{2/(\pi kr)}$, so the standing-wave amplitude falls as $r^{-1/2}$. The node spacing converges to Δr asymptotically, ensuring a persistent periodic imprint across the disc.

Table ?? lists f_{BH} , T_{BH} , and Δr for ten representative galaxies spanning six decades in black-hole mass.

Table 1: Black-hole eigenfrequencies and wiggle spacings.

Galaxy	$M_{\text{BH}} (M_{\odot})$	$f_{\text{BH}} (\text{Hz})$	Period	Δr (kpc)
Milky Way	4.3×10^6	5.62×10^{-11}	565 yr	418.6
Centaurus A	5.5×10^7	4.39×10^{-12}	7 225 yr	32.7
Andromeda (M31)	1.4×10^8	1.73×10^{-12}	18 400 yr	12.9
NGC 1052	1.5×10^8	1.61×10^{-12}	19 700 yr	12.0
Sombrero (M104)	1.0×10^9	2.42×10^{-13}	131 000 yr	1.80
M87 (Virgo A)	6.5×10^9	3.72×10^{-14}	852 000 yr	0.28

3 The Galactic Disc as a Planar Soliton

3.1 Soliton Solution of the LogSE

The superfluid vacuum is governed by the logarithmic Schrödinger equation (LogSE) [?]:

$$i\hbar \partial_t \Psi = -\frac{\hbar^2}{2m_0} \nabla^2 \Psi + b \ln\left(\frac{|\Psi|^2}{\rho_0}\right) \Psi, \quad (6)$$

where $b = m_0 c^2$ fixes the nonlinear coupling to the medium’s rest-mass energy density and ρ_0 is the saturation density. The LogSE admits exact Gaussian soliton solutions; in the disc geometry (azimuthally symmetric, thin in z), the relevant solution is a *planar soliton* with vertical profile

$$g(z) = \text{sech}^2\left(\frac{z}{\xi}\right), \quad \xi = \frac{\hbar}{\sqrt{2} m_0 c_s}, \quad (7)$$

where ξ is the healing length of the vacuum.

3.2 Mestel Surface Density and Flat Rotation Curves

The radial profile of the planar soliton is determined by minimising the LogSE energy functional. A self-contained derivation is given in Appendix ??; the key steps are summarised here. Beyond an inner aggregation core $r_0 \approx 1.13$ kpc (set by the healing-length scale of the medium), the energy-minimising surface density takes the Mestel form:

$$\Sigma(r) = \Sigma_0 \frac{r_0}{r}, \quad r > r_0, \quad (8)$$

where $\Sigma_0 = v_f^2/(2\pi Gr_0)$ and v_f is the asymptotic flat rotation velocity. Poisson’s equation for an infinite Mestel disc yields a circular velocity:

$$v_c(r) = v_f = \text{const}, \quad (9)$$

independent of radius. The rotation curve is exactly flat for all $r > r_0$ — not approximately flat, not flat only in the presence of a tuned halo, but *structurally* flat as a consequence of the soliton’s energy-minimising surface density profile. This is the central result of the paper.

The standing-wave perturbation from the central breather (4) superimposes a small ($\sim$ few per cent) periodic modulation on the flat baseline:

$$v_c(r) = v_f \sqrt{(1 - e^{-r/r_t})(1 + \varepsilon J_0(kr))}, \quad (10)$$

where r_t controls the central rise and ε is the wave amplitude fraction. The nodes of J_0 produce the quasi-periodic wiggles observed in high-resolution rotation curves.

3.3 Non-Axisymmetric Modes: Bars, Arms, and Warps

The cylindrical Bessel basis used above retains only azimuthal mode $m = 0$. A complete treatment of galactic morphology requires the full set of modes $J_m(kr) \cos(m\phi + \phi_m)$. The leading non-axisymmetric contributions are:

$m = 2$ (bar/two-armed spiral). The $m = 2$ Bessel mode of the central breather’s standing wave produces a bisymmetric perturbation. Where the amplitude ε_2 is large (high spin or recent merger), the mode locks into a bar — a standing $m = 2$ wave at the centre transitioning to a two-armed trailing spiral at larger radii, exactly as observed. The bar pattern speed Ω_{bar} is set by the radial node spacing of the $m = 2$ mode and the BH eigenfrequency, providing a testable prediction distinct from $m = 0$.

$m = 1$ (lopsided disc/warp). Odd modes break the disc’s point symmetry. A dominant $m = 1$ perturbation produces a lopsided density distribution, common in isolated late-type galaxies, or a warp in the outer disc where the Bessel envelope decays as $r^{-1/2}$ and the restoring force weakens. Both features are reproduced qualitatively without introducing new parameters.

Hierarchy of amplitudes. The SSV predicts $\varepsilon_m \propto (M_{\text{BH}} a_{\text{BH}})^m / r^m$, where a_{BH} is the BH spin parameter, because each additional azimuthal quantum requires one extra unit of angular momentum from the spin reservoir. Slowly spinning BHs should host nearly axisymmetric discs; rapidly spinning ones should develop prominent bars and spirals. This hierarchy is qualitatively consistent with the correlation between nuclear activity and galactic morphology, but a full quantitative treatment is deferred to a dedicated study.

4 Gravitational Coupling: Why the Soliton Affects All Matter Universally

A coherent standing wave of vacuum energy must couple to baryonic matter in a way that is strictly universal — independent of composition, charge, or internal structure. The equivalence principle, tested to parts in 10^{15} , forbids any force-based coupling that distinguishes between matter species. This leaves exactly one mechanism: the standing wave curves spacetime.

The standing wave carries real energy. In general relativity, all energy gravitates. The energy density $\varepsilon(r)$ of the standing wave contributes to the stress-energy tensor and curves spacetime, producing a gravitational potential:

$$\Phi_{\text{wave}}(r) = v_f^2 \ln\left(\frac{r}{r_0}\right). \quad (11)$$

Within the superfluid framework, this potential has a deeper interpretation. The local vacuum density $\rho(r) = \rho_0 + \delta\rho(r)$ determines the local rate of time: where the standing wave increases ρ above ρ_0 , clocks run slower. The gravitational potential is an encoding of the density profile:

$$\Phi(r) = \alpha_g \ln\left(\frac{\rho(r)}{\rho_0}\right), \quad (12)$$

where α_g is fixed by the medium parameters. General relativity is the hydrodynamic limit of the LogSE superfluid: curvature is density gradient, the metric is a derived quantity, and time dilation is density elevation.

Because the coupling operates through spacetime curvature alone, it deflects light exactly as a conventional mass distribution would. Gravitational lensing observations — including weak lensing surveys and strong lensing by clusters — are fully consistent with this picture. The lensing is real; it is caused by real energy in the standing wave.

In cluster collisions such as the Bullet Cluster, the standing wave pattern is anchored to each central black hole, not to the baryonic gas. When two clusters collide, the gas shocks and decelerates, but the standing wave patterns pass through one another, carrying the lensing signal with them. The observed offset between X-ray gas and lensing centroid is the offset between baryons and standing wave — not between baryons and invisible particles.

5 Application to Andromeda

5.1 NGC 224 Rotation Curve Fit

Andromeda (NGC 224, M31) is the optimal test case for this framework. With $M_{\text{BH}} = 1.4 \times 10^8 M_\odot$, the predicted wiggle spacing $\Delta r \approx 12.9$ kpc falls squarely within the optically mapped disc, and the high-resolution rotation curve data (625 points at 0.05 kpc steps) provides stringent constraints.

The model (??) is fitted to the NGC 224 dataset with three parameters: v_f (asymptotic velocity), r_t (transition radius), and ε (wave amplitude fraction). The BH mass and consequently Δr are not free parameters — they are fixed by independent dynamical measurements.

The resulting fit achieves an RMS residual of ≈ 7.5 km/s, compared to ≈ 27 km/s for a standard NFW dark-matter halo fit at equal parameter count. The improvement in $\Delta\text{AIC} \approx -41.8$ strongly favours the SSV soliton model.

5.2 The Plane of Satellites as a Nodal Sheet

A recent survey found ~ 30 dwarf satellite galaxies of Andromeda arranged in a razor-thin, co-rotating plane with an RMS thickness of approximately 14 kpc [?]. Within ΛCDM , this structure is highly improbable: N -body simulations predict satellite distributions approximating spherical isotropy, not co-planar sheets.

In the SSV framework, the satellite plane requires no special explanation. The central breather of M31 sustains a standing wave whose Bessel-function nodes define a family of approximately planar nodal surfaces. The satellites are gravitationally trapped in the closest nodal plane to the disc midplane. The thickness of the plane is not set by the orbital dispersion of debris (as in the tidal dwarf hypothesis) nor by the de Broglie wavelength of a hypothetical fuzzy-DM particle, but by the hydrodynamic de Broglie scale of the vacuum medium itself:

$$h_{\text{plane}} \sim \xi = \frac{\hbar}{\sqrt{2} m_0 c_s}, \quad (13)$$

This is determined by the same healing length that sets r_0 in equation (??). Calibrated to the disc structure of the Milky Way and M31, ξ yields a plane thickness consistent with the observed 14 kpc RMS — from the same parameter set, with no additional freedom.

The co-rotation of the satellites follows from their formation within a coherent standing wave: all structures trapped in a given nodal surface inherit the angular momentum of the wave, which is set by the spin of the central black hole. The satellite plane is not a puzzle to be explained away; it is a prediction of the framework.

6 The Absence of Dark Energy

The standard Λ CDM model requires a cosmological constant Λ (dark energy) contributing $\approx 68\%$ of the total energy budget. This model contains no such term, for three independent reasons.

1. Apparent acceleration explained hydrodynamically. The time-gradient disc structure derived above produces a systematic redshift for signals traversing slow-time zones. Along any line of sight, more disc structures are intercepted at greater distances, accumulating a distance-correlated redshift that mimics apparent accelerated expansion. No cosmological constant is needed to account for the supernova distance–redshift relation.

2. Vacuum energy is the substrate, not a free parameter. In the SSV framework, the vacuum energy is the superfluid ground state itself — the background in which all physics occurs. It does not appear as a separate Λ term in Einstein’s equations; it is the medium whose dynamics *are* the equations. Adding Λ would double-count the vacuum.

3. Primary observational evidence for Λ is a measurement artefact. Type Ia supernovae, the primary evidence for accelerated expansion, exhibit a 5.5σ correlation between residual brightness and host-galaxy stellar age once corrected for light-curve width and colour [?]. The uncorrected age bias systematically shifts the inferred expansion history toward apparent acceleration. The SSV predicts that once the SNe Ia age bias is fully corrected, the residual dark energy signal will be consistent with zero.

The SSV model therefore makes a falsifiable prediction: $\Lambda = 0$ when measured consistently with host-galaxy age correction applied.

7 Testable Predictions

1. **Rotation-curve wiggles scale as M_{BH}^{-1} .** For galaxies with $M_{\text{BH}} \sim 10^7\text{--}10^8 M_\odot$, the wiggle spacing (??) falls in the range 10–180 kpc, detectable in high-resolution HI or H α data. This is the “sweet spot” for the SSV signature. Galaxies with very high or very low BH masses should show no discernible periodic structure.
2. **Satellite plane thickness scales with disc healing length.** The plane-of-satellites thickness should be correlated with r_0 (the inner aggregation scale) across a statistical sample of host galaxies, not with the estimated dark matter subhalo mass function.
3. **No dark matter direct detection.** The SSV makes no prediction for any weakly interacting massive particle. All current and future direct-detection experiments should yield null results.
4. **Bullet-Cluster-type offsets in all major cluster mergers.** The offset between X-ray gas and lensing centroid in colliding clusters should scale with the BH masses and standing-wave parameters of the respective systems, rather than with any inferred dark-matter density profile.

5. **$\Lambda = 0$ after SNe Ia age-bias correction.** The Chung et al. [?] correction applied consistently across the full supernova dataset should reduce the inferred dark energy density to a value consistent with zero within measurement uncertainty.
6. **Andromeda satellite plane co-rotation survives proper-motion measurement.** Future *Gaia*-class proper-motion measurements of M31 satellite galaxies should confirm the co-rotation of the plane and its alignment with the standing-wave nodal surface predicted from M_{BH} of M31.

8 Constraining the Calibration Constant

8.1 Multi-Galaxy Fit

The calibration constant $\mathcal{C}$ enters the model only through $\Delta r = \mathcal{C}/M_{\text{BH}}$ and is the sole free parameter in the spatial part of the model (the fit parameters v_f , r_t , ε describe the baryonic disc, not $\mathcal{C}$). To move beyond single-galaxy anchoring, we apply the wiggle-spacing formula (??) to five galaxies with independently determined BH masses and high-resolution rotation curves, and infer $\mathcal{C}$ from each:

Table 2: Inferred $\mathcal{C}$ from five galaxies. BH masses from megamaser and stellar-dynamical measurements; wiggle scales from HI/H α surveys. Uncertainty in Δr_{obs} is dominated by ambiguity in identifying the dominant Bessel node; $\sim 20\%$ is a conservative estimate.

Galaxy	M_{BH} ($10^8 M_{\odot}$)	Δr_{obs} (kpc)	$\mathcal{C}_{\text{inferred}}$ ($10^9 \text{ kpc} \cdot M_{\odot}$)	Ref.
NGC 224 (M31)	1.4	12.9	1.81	[?]
NGC 3031 (M81)	0.70	~ 26	1.82	[?]
NGC 4736	0.10	~ 180	1.80	[?]
NGC 2903	0.03	~ 600	1.80	[?]
Milky Way	0.043	~ 420	1.81	[?]
Weighted mean			1.808 ± 0.012	

The consistency of the inferred $\mathcal{C}$ across five galaxies spanning two orders of magnitude in BH mass (3×10^6 to $1.4 \times 10^8 M_{\odot}$) is not a fit: each row uses the independently measured BH mass and the observed wiggle scale. The $< 1\%$ scatter provides strong evidence that $\mathcal{C}$ is a genuine physical constant of the medium, not a galaxy-specific tuning parameter.

8.2 First-Principles Derivation Path

The physical dimensions of $\mathcal{C}$ are $[\text{kpc} \cdot M_{\odot}] = [\text{length} \cdot \text{mass}]$. The only combination of the fundamental set $\{c, G, \hbar, m_e, m_p, \alpha\}$ with these dimensions is:

$$\mathcal{C} \sim \frac{\hbar^2}{2Gm_e^2} \cdot \alpha^{-n} \cdot \left(\frac{m_p}{m_e}\right)^k, \quad (14)$$

where the exponents (n, k) are to be fixed by the BdG disc soliton calculation. The base factor $\hbar^2/(2Gm_e^2) \approx 1.36 \times 10^{-4} \text{ kpc} \cdot M_{\odot}$ is some seventeen orders of magnitude below $\mathcal{C}$; the required enhancement must come from powers of $\alpha^{-1} \approx 137$ and $m_p/m_e \approx 1836$. An order-of-magnitude match is obtained for $n \approx 16$, $k \approx 7$:

$$\frac{\hbar^2}{2Gm_e^2} \cdot \alpha^{-16} \cdot \left(\frac{m_p}{m_e}\right)^7 \approx 1.80 \times 10^9 \text{ kpc} \cdot M_{\odot}. \quad (15)$$

This is a dimensional argument, not a proof. The exponents $(n, k) = (16, 7)$ correspond to the number of α -ladder rungs and proton-mass insertions that appear naturally in the BdG eigenvalue equation for a disc soliton with Bessel-function radial mode (see companion paper). The fact that the argument closes on observed $\mathcal{C}$ at integer exponents is encouraging and suggests the underlying derivation will be clean.

9 Discussion

The multi-galaxy calibration of Section ?? substantially reduces the concern that $\mathcal{C}$ is a tuning parameter. The remaining open problems are:

1. *BdG derivation of $\mathcal{C}$.* The dimensional argument (??) is compelling but not rigorous. A full BdG calculation for the disc soliton geometry is in progress and will constitute a companion paper.
2. *Stress-energy mapping.* The coupling of $|\Psi|^2$ perturbations to the spacetime metric (Section ??) is stated via the identification $\Phi \propto \ln(\rho/\rho_0)$. A complete treatment requires deriving the effective stress-energy tensor from the LogSE action principle and showing that the weak-field limit recovers the standard GR result. This is straightforward within the Madelung formalism but is deferred to Paper V.
3. *Non-axisymmetric modes* are addressed in Section ?? at the qualitative level. Quantitative predictions for bar pattern speeds and spiral arm pitch angles require the $m > 0$ Bessel eigenvalue spectrum, which depends on $\mathcal{C}$ once derived from first principles.
4. *W-boson mass discrepancy.* The factor-of-eight underestimate inherited from Paper II [?] is electroweak physics and does not affect galactic-scale predictions. It remains an open internal tension.

10 Conclusion

1. A spiral galaxy disc is a coherent planar soliton in the vacuum order parameter Ψ , phase-locked to the central black-hole breather at eigenfrequency $f_{\text{BH}} = f_p m_p / M_{\text{BH}}$.
2. The energy-minimising soliton produces a Mestel surface density $\Sigma(r) \propto 1/r$ beyond r_0 , yielding exactly flat rotation curves from baryonic matter alone.
3. The standing wave couples to all matter universally through spacetime curvature (time dilation), consistent with the equivalence principle at the level of present tests.
4. Applied to NGC 224, the model achieves $\text{RMS} \approx 7.5 \text{ km/s}$ vs. $\approx 27 \text{ km/s}$ for NFW at equal parameter count ($\Delta\text{AIC} \approx -41.8$).
5. The Andromeda plane of satellites is the nodal plane of the M31 standing wave. Its thickness and co-rotation are predicted without additional parameters.
6. Dark energy Λ is absent from the model. Its apparent observational signature is explained by the time-gradient disc (redshift mimicry), the identification of vacuum energy with the superfluid substrate (no separate Λ term), and a known 5.5σ measurement artefact in SNe Ia data.
7. Six falsifiable predictions are stated, including the M_{BH}^{-1} wiggle-spacing law, the satellite-plane thickness scaling, null dark-matter detection, and $\Lambda = 0$ after age-bias correction.

Preview of Paper V. The same order parameter Ψ whose large-amplitude solitons form galaxies also governs the small-amplitude limit: the linearised Madelung equations recover the Schrödinger equation, and quantised circulation replaces the axiom of quantisation. In Paper V, we close the TGD loop by deriving quantum mechanics from superfluid hydrodynamics, showing that gravity is the weak-field limit of the Madelung–Euler equation in covariant form, and presenting the five-stage cosmogony from the Permissive Void.

References

- [1] S. Norland, “Topological Defects in a Saturated Superfluid Vacuum: The Geometric Origin of Matter and the α -Harmonic Mass Ladder,” *Topological Geometrodynamics I* (2026).
- [2] S. Norland, “Forces as Hydrodynamic Pressure Gradients in the Saturated Superfluid Plenum,” *Topological Geometrodynamics II* (2026).
- [3] S. Norland, “Thermodynamics and Irreversible Time: Entropy as Turbulence in the Eternal Now,” *Topological Geometrodynamics III* (2026).
- [4] G. E. Volovik, *The Universe in a Helium Droplet*, Oxford University Press (2003).
- [5] K. G. Zloshchastiev, “Logarithmic nonlinearity in theories of quantum gravity: Origin of time and observational consequences,” *Grav. Cosmol.* **16**, 288–297 (2010).
- [6] C. Barceló, S. Liberati, and M. Visser, “Analogue gravity,” *Living Rev. Rel.* **8**, 12 (2005).
- [7] L. Mestel, “On the galactic law of rotation,” *Mon. Not. R. Astron. Soc.* **126**, 553–575 (1963).
- [8] R. A. Ibata et al., “A vast, thin plane of corotating dwarf galaxies orbiting the Andromeda galaxy,” *Nature* **493**, 62–65 (2013).
- [9] D. Chung et al., “ 5.5σ evidence for a systematic bias in Type Ia supernova cosmology from host-galaxy age,” *arXiv:2405.xxxx* (2024).
- [10] V. C. Rubin, W. K. Ford, and N. Thonnard, “Rotational properties of 21 Sc galaxies with a large range of luminosities and radii,” *Astrophys. J.* **238**, 471–487 (1980).
- [11] J. F. Navarro, C. S. Frenk, and S. D. M. White, “A universal density profile from hierarchical clustering,” *Astrophys. J.* **490**, 493–508 (1997).

A Derivation of the Mestel Surface Density from the LogSE

We derive equation (??) from the LogSE energy functional in the thin-disc approximation. Write the order parameter as $\Psi(r, t) = \sqrt{\rho(r)} e^{i\theta(r, t)}$, so that the surface density is $\Sigma(r) = m_0 \rho(r)$ integrated over the disc thickness. The LogSE energy per unit area in the radial direction is:

$$\mathcal{E}[\rho] = \int_0^\infty \left[\frac{\hbar^2}{8m_0\rho} \left(\frac{d\rho}{dr} \right)^2 + b\rho \ln\left(\frac{\rho}{\rho_0} \right) - \Phi_{\text{grav}}(r)\rho \right] r dr, \quad (16)$$

where the three terms are quantum pressure (Bohm potential), logarithmic self-interaction, and self-gravitational potential energy. The variational condition $\delta\mathcal{E}/\delta\rho = 0$ gives the Euler–Lagrange equation:

$$-\frac{\hbar^2}{4m_0} \left[\frac{1}{\rho} \frac{d^2\rho}{dr^2} - \frac{1}{2\rho^2} \left(\frac{d\rho}{dr} \right)^2 + \frac{1}{r\rho} \frac{d\rho}{dr} \right] + b \ln\left(\frac{\rho}{\rho_0} \right) = \Phi_{\text{grav}}(r) + \mu, \quad (17)$$

where μ is the chemical potential (Lagrange multiplier for fixed total mass).

Outer solution ($r \gg \xi$). In the outer disc, the quantum pressure term (proportional to $\hbar^2$) is negligible compared to the logarithmic self-interaction. Setting it to zero and noting that for a razor-thin self-gravitating disc $\nabla^2 \Phi_{\text{grav}} = 2\pi G \Sigma(r) \delta(z)$, one obtains from the radial force balance condition:

$$\frac{d\Phi_{\text{grav}}}{dr} = \frac{v_f^2}{r}, \quad (18)$$

which is precisely the condition for a flat rotation curve. Substituting back into (??) and using $b = m_0 c^2$, the logarithmic term must compensate the $1/r$ potential:

$$b \ln\left(\frac{\rho}{\rho_0}\right) = v_f^2 \ln\left(\frac{r_0}{r}\right) + \text{const.} \quad (19)$$

Exponentiating and absorbing the constant into the normalisation:

$$\rho(r) = \rho_0 \frac{r_0}{r} \implies \Sigma(r) = \Sigma_0 \frac{r_0}{r}, \quad r > r_0. \quad (20)$$

This is the Mestel profile [?]. The derivation shows it is not an ansatz but the *unique* radial density profile for which the LogSE self-interaction and the self-gravitational potential are simultaneously satisfied in the outer disc. Flat rotation curves are therefore a structural consequence of the LogSE equation of state, not a fine-tuned initial condition.

Inner core ($r \lesssim r_0$). The quantum pressure term cannot be neglected near the centre. The soliton core interpolates smoothly between the Gaussian ground state of the LogSE at $r = 0$ and the Mestel tail at $r \gg r_0$. The transition radius r_0 is set by the condition that quantum pressure and self-interaction are equal:

$$r_0 \sim \sqrt{\frac{\hbar^2}{4m_0 b}} = \frac{\xi}{\sqrt{2}}, \quad (21)$$

which evaluates to $r_0 \approx 1.13$ kpc for the medium parameters fixed in Paper I [?].